

Integrated Enterprise Network Capstone

Segmentation, Centralized Services, Authentication, Monitoring, and Access-Layer Hardening — Built to One Standard and Proven to Interoperate for a Growing Financial-Services Firm

Certification track: Cisco Certified Network Associate (CCNA 200-301) — Network Access, IP Connectivity, IP Services, and Security Fundamentals, with CyberOps 200-201 security concepts as secondary alignment

Simulation platform: Cisco Packet Tracer 8.2 or later (no other software required)

Team size and effort: 4 students × 12 hours per week × 8 weeks = 384 total person-hours

PROJECT SCOPE DOCUMENT	
Project Name:	Sponsor:
Integrated Enterprise Network Capstone Segmentation, Centralized Services, Authentication, Monitoring, and Access-Layer Hardening — Built to One Standard and Proven to Interoperate for a Growing Financial-Services Firm	Dr. Osama Eljabiri
Business Unit Project Leader:	Project Manager:
Mr. Richard Thompson	Benjamin Popso
Requested Start Date:	Requested End Date:
September 29, 2026	November 24, 2026
Objective:	
The goal of this project is to design, build, and validate a complete enterprise network for Kestrel Financial Partners — a growing financial-services firm moving from an unmanaged flat office network to an internally administered enterprise — integrating departmental segmentation, centralized address and name services, centralized administrative authentication, inter-segment access control, external and wireless connectivity, and centralized monitoring into one environment, deployed to a single documented build standard rather than improvised per device, and proven to interoperate under induced failure. Unlike the single-topic scopes in this series, the graded core of this project is not any one technology but the discipline of making all of them work together correctly and consistently.	
Narrative:	
Kestrel Financial Partners has outgrown its original flat, loosely managed network. As headcount has expanded across IT and Admin, Finance, General Staff, and server infrastructure, the network has not kept pace: there is no meaningful separation between departments, no centralized way to authenticate administrators, no consistent access control between sensitive financial resources and general-staff traffic, and no centralized visibility into what is happening on the network. The firm needs an enterprise network built the way a real financial-services company at this stage would build one — segmented, authenticated, access-controlled, and monitored — sized for its size and risk profile rather than for a large regulated enterprise. This is deliberately an integration project, and that is what distinguishes it from every other scope in the series. The other projects go deep on a single technology — controller-based wireless, role-based identity and accounting, security operations and correlation, zone-based ACL design with exhaustive verification. This one does the opposite: it deploys each of those areas to competent, functional CCNA depth and spends its rigor on the far harder	

problem of making independently built components coexist. A network where wireless works, RADIUS works, DHCP works, and the ACLs work — but where the ACL silently blocks RADIUS, or the wireless clients never get a DHCP address, or the Syslog timestamps do not line up because NTP was an afterthought — is not a working network. The integration failures are the intellectual content.

The team builds a segmented topology reflecting the firm's departmental structure — VLANs for IT and Admin, Finance, General Staff, Servers, Management, and Wireless users — with trunking and inter-VLAN routing between them. It then layers in the services and controls a growing firm needs to operate that network safely: DHCP and DNS for centralized addressing and name resolution; RADIUS and AAA for centralized administrative authentication with SSH replacing unencrypted management access; extended ACLs enforcing the access relationships between segments; NAT and PAT for outbound external connectivity; Syslog and NTP for centralized, time-correct logging; secure wireless connectivity to an access point; and access-layer hardening through port security, PortFast, BPDU Guard, and unused-port protection.

Every component is deployed to a single build standard — a documented, consistent configuration applied identically across every device of a kind, not reinvented switch by switch. The build standard is itself a deliverable, because consistency is what makes a heterogeneous enterprise maintainable and is precisely the discipline this project exists to develop. A device that works but does not conform to the standard is treated as a defect, not a success.

Validation is where the project earns its rigor. Beyond confirming each component works in isolation, the team proves the components interoperate through cross-service tests that exercise several at once — a wireless client that must associate, authenticate against RADIUS, receive a DHCP address, resolve a name via DNS, be permitted to its own segment's resources and denied others by ACL, reach the internet through NAT, and appear in the Syslog with an NTP-correct timestamp, all in one journey. The team then induces failure deliberately: break NTP and show the Syslog timeline decorrelate; misorder an ACL and show it silently break RADIUS or DHCP; then diagnose and repair each, documenting the integration conflict and its resolution. Finally, a team member who did not build a given component audits it against the build standard, so conformance is verified independently rather than self-attested. The deliverable is a complete enterprise network that has been proven not merely to have all its parts, but to work as one coherent, consistent, resilient whole.

Risks:

What are the potential challenges that this project may face?

RADIUS and AAA integration takes longer than expected

Impact: Centralized authentication is typically the least familiar technology to the team and has the most interdependent pieces — the RADIUS server, the network-access-server client configuration, and the authentication method — making it the most likely place for the schedule to slip and the most likely to conflict with the ACL layer built alongside it. **Mitigation:** RADIUS and AAA are scheduled in Week 4, ahead of the access-control layer that most commonly breaks them, with a defined fallback of local authentication plus a documented explanation of the intended RADIUS design if full functional integration cannot be completed. Cross-service test T-11 specifically verifies that the ACL layer does not block RADIUS.

Integration conflicts between independently built components

Impact: With four members building different services in parallel, a component that works in isolation may fail once integrated — an ACL blocking legitimate RADIUS or DHCP traffic, a wireless VLAN with no DHCP scope, Syslog entries with wrong timestamps because NTP was not yet configured. These are the project's characteristic failures and the reason it exists. **Mitigation:** The addressing and segmentation plan and the build standard are fixed in Week 1 so all members build against the same reference. A dedicated integration and induced-failure week (Week 7) precedes delivery, and the cross-service test matrix in Appendix F is designed to surface exactly these conflicts rather than test components in isolation.

Component depth creeping into sibling-scope territory

Impact: A team that over-invests in one component — building controller-grade wireless or role-based accounting — spends its hours where another scope already owns the depth, and under-delivers the integration that is this

project's actual graded core. **Mitigation:** The out-of-scope list explicitly caps each component at functional CCNA depth and points to the sibling scope that owns the deep version. The rubric weights integration at 25%, more than any single component, so effort is steered toward interoperability rather than depth in one area.

Build-standard drift across devices

Impact: Without a consistent standard, each device is configured slightly differently, the environment becomes unmanageable, and the non-author audit cannot produce a meaningful pass/fail because there is nothing consistent to audit against. **Mitigation:** The build standard is a Week 1 deliverable and the single source of truth; the Week 7 non-author audit checks every device against it, and a conforming-but-different device is a defect. Configurations are committed to version control every session so drift is visible in the history.

Packet Tracer limitations on a specific feature

Impact: A specific configuration — a RADIUS attribute, a wireless security mode, a Syslog or NTP behaviour — may differ from or be unavailable in Packet Tracer relative to production IOS. **Mitigation:** A Week 2 capability spike confirms the required feature set. Any feature found unsupported is documented as analyzed rather than demonstrated, with an available compensating configuration deployed in its place and the production behaviour explained.

Uneven contribution across team members

Impact: A division of work not followed in practice leaves gaps in the final deliverable and an inaccurate picture of individual contribution — and, because this is an integration project, a gap in one member's component breaks the whole. **Mitigation:** Appendix C assigns clear primary accountability per member, with shared responsibility for integration; the Week 5 cross-training rotation and the Week 7 non-author audit both require members to work outside their own component; and the repository commit history must reflect contribution from all four.

Dependencies:

What are the tools, people, technical conditions or software/platforms that the project depends on?

- Cisco Packet Tracer 8.2 or later installed on all four team members' machines at an identical version
- Cisco Networking Academy access for CCNA 200-301 material spanning VLANs, IP services, AAA, wireless, ACLs, and access-layer hardening
- A shared version control repository holding the .pkt file, the build standard, exported configurations, and documentation
- A diagramming tool for the logical and physical topology deliverables — draw.io, Lucidchart, or Microsoft Visio
- A spreadsheet tool for the addressing and segmentation plan, the build standard checklist, and the acceptance test matrix
- A weekly team meeting of no less than two hours with all four members present
- Sponsor design review at the end of Week 1 to approve the addressing and segmentation plan and the build standard before component build begins

Boundaries: (In Scope / Out of Scope)

What requirements/features are considered part of the scope to consider the project complete?

IN SCOPE

- Departmental segmentation via VLANs for IT and Admin, Finance, General Staff, Servers, Management, and Wireless users, with trunking and inter-VLAN routing between all segments
- A VLSM IPv4 addressing and subnetting plan covering every segment, fixed in Week 1 as the shared reference for all members
- A documented build standard — a consistent configuration applied identically across every device of a kind — as a first-class deliverable
- DHCP providing correct per-segment addressing, and DNS supporting internal name resolution

- Centralized AAA authentication for network-device administration using RADIUS, deployed and functionally tested, with SSH replacing unencrypted management access
- Extended ACLs enforcing the access relationships between segments, verified in both the permitted and the denied direction
- NAT and PAT for outbound external connectivity
- Syslog collection and NTP time synchronization for centralized, time-correct logging
- Basic secure wireless connectivity to an access point using a standard secure authentication method, integrated with the segment's DHCP and ACL policy
- Port security, PortFast, BPDU Guard, and unused-port protection on access-layer switches
- Cross-service interoperation testing that exercises multiple components in a single end-to-end journey
- Induced-failure integration testing — deliberately breaking a dependency (e.g. NTP, or ACL ordering) and demonstrating the resulting integration failure and its repair
- A non-author audit of each component against the build standard, producing a pass/fail per component
- Full device configuration export and logical and physical topology diagrams

What requirements are considered part of the next phase, the work of other teams, or not included in what is expected for this team to do?

OUT OF SCOPE

- Controller-based wireless, RF channel and power planning, roaming, or WPA2-Enterprise design depth — this project deploys basic secure wireless to a single access point only; the deep wireless design is owned by the dedicated wireless scope (CCNA-03)
- Role-based command authorization, per-command accounting, accountability reconstruction, or fallback-path design — this project uses RADIUS for administrative login only; the deep identity and AAA design is owned by the dedicated identity scopes (CYBEROPS-10 / CYBEROPS-11)
- Security monitoring, log correlation, triage, playbooks, or incident response — this project centralizes logging so events are collected and time-correct; the SOC and monitoring depth is owned by the dedicated monitoring scope (CYBEROPS-07)
- Zone-based ACL design with an exhaustive directional verification matrix and per-service granularity — this project enforces and verifies inter-segment ACLs at functional depth; the deep segmentation design is owned by the dedicated segmentation scope (CCNA-12)
- Multi-area OSPF engineering, summarization design, or convergence tuning — inter-VLAN routing here is functional; deep routing design is owned by the dedicated routing scope (CCNA-01)
- Formal adversarial or penetration testing of the completed environment
- High-availability or redundancy design, including redundant links and first-hop redundancy protocols
- SIEM or advanced log analysis beyond centralized Syslog collection
- User-facing directory services or 802.1X port-based network access control for end hosts
- Physical hardware, IPv6, WAN or site-to-site VPN, QoS, and any deployment outside the Packet Tracer simulation environment
- Ongoing operational support, monitoring, or incident response after project handover

Project Deliverables:

What are the key deliverables that are expected from the team to submit by the end of the semester?

D1. Cisco Packet Tracer simulation file (.pkt)

The complete integrated enterprise network with every component deployed, opening without error in Packet Tracer 8.2 or later, with every device named per the build standard and all configurations saved to startup-config.

D2. Network topology and addressing design

Logical and physical topology diagrams and the VLSM IPv4 addressing and subnetting plan covering every defined segment, fixed in Week 1 as the shared build reference.

D3. Build standard

The documented configuration standard applied identically across every device of a kind — naming, management access, hardening baseline, service settings — against which the Week 7 non-author audit is run. Consistency is the point: a working-but-nonconforming device is a defect. The standard itself can be modified and upgraded as the team learns and implements the build; what matters is that it stays consistent going forward, so the same technology carries the same configuration and a device is not flagged as nonconforming simply because the standard was refined after it was built.

D4. Segmented and routed network

VLANs, 802.1Q trunking, and inter-VLAN routing deployed and functioning across all departmental segments.

D5. Centralized services

DHCP and DNS deployed and functioning across the appropriate segments; NAT and PAT providing outbound external connectivity; Syslog and NTP providing centralized, time-correct logging.

D6. Authentication and secure administration

RADIUS and AAA authentication for device administration and SSH-enforced management access, deployed and functionally tested, with the fallback design documented if full RADIUS integration cannot be completed.

D7. Access control, wireless, and access-layer hardening

Extended ACLs enforcing the inter-segment access relationships verified in both directions; secure wireless to an access point integrated with its segment's DHCP and ACL policy; and port security, PortFast, BPDU Guard, and unused-port protection on access-layer switches.

D8. Integration and interoperation evidence

The cross-service test results, the induced-failure integration tests with their diagnosis and repair, and the documented log of integration conflicts found and resolved — the distinctive core evidence of this project.

D9. Non-author audit results

A pass/fail-per-component audit of the build against the build standard, performed by team members auditing components they did not build.

D10. Exported configurations, final report, presentation, and video demonstration

Plain-text running configurations for every device, the consolidated report, the Week 8 slide deck, and the recorded demonstration as specified in Appendix I.

Project Assumptions:

What standards/policies should be followed?

- Standard Cisco networking and security practices at CCNA level inform the configuration of each component
- Every component is deployed to a single documented build standard applied identically across every device of a kind; a working-but-nonconforming device is a defect
- Every component is functionally demonstrated and, where it depends on another, demonstrated in integration; a component shown working only in isolation is incomplete
- Every access relationship is proven in both the permitted and the denied direction; a boundary asserted but not tested in the denied direction is unproven
- Where a component cannot be demonstrated in Packet Tracer, it is documented as analyzed rather than demonstrated with a compensating configuration in its place

What are the things that the sponsor assumes the team should have or provide?

- A functioning integrated enterprise network demonstrating every component in Boundaries, working together
- Cross-service and induced-failure integration evidence, not only per-component functional evidence
- A non-author audit of each component against the build standard
- A video demonstration showing the cross-service end-to-end journey and at least one induced-failure repair
- Attendance by all four members at the final presentation, each able to explain the complete environment, not only their own components
- A repository commit history showing genuine contribution from all four members

What are the things that the team assumes the sponsor should provide?

- A design review at the end of Week 1 to approve the addressing and segmentation plan and the build standard before component build begins
- The grading rubric in advance, as supplied in Appendix G
- Access to relevant course material
- Escalation access to faculty when the team is blocked for more than one working session

Estimated Time (and Budget):

The project will take 8 weeks to be completed, at 12 hours per week per student across a team of 4 students, for a total of 384 person-hours — matching the effort baseline of the approved capstone scope series. The Work Breakdown Structure in Appendix B allocates 48 person-hours per week and defines a completion gate for each week. Because this is an integration project, the hours are weighted toward Weeks 5–7, where components meet. Budget: \$0. Cisco Packet Tracer 8.2 or later and Cisco Networking Academy course material are provided at no cost; no hardware, cloud subscription, or commercial software licence is required.

Budget: \$0. All tooling is free of charge — Cisco Packet Tracer 8.2 or later and Cisco Networking Academy course material are provided at no cost to enrolled students. No hardware purchase, cloud subscription, or commercial software licence is required to complete any deliverable in this scope.

Signatures

Sponsor Signature

See Below

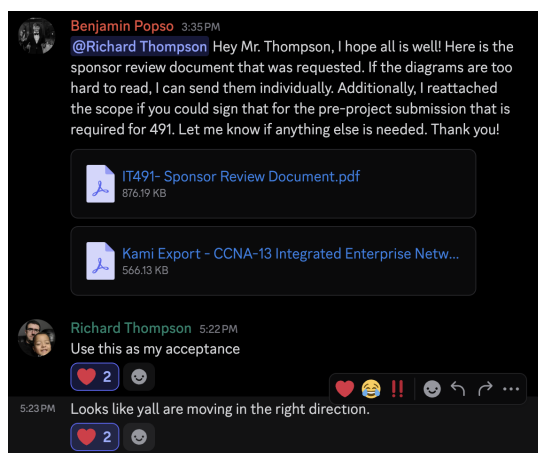
Date: 10/4/2026

Project Manager Signature

Benjamin Popso

Benjamin Popso

Date: September 27, 2026



Appendix A — Cisco Certification and NetAcad Alignment

This project touches more CCNA 200-301 domains than any single scope in the series, by design — integration is its subject, so it spans Network Access, IP Connectivity, IP Services, and Security Fundamentals at once, with CyberOps 200-201 Security Concepts as secondary alignment where the layered controls and centralized logging apply. Each activity is tied to the NetAcad module that develops it. The intent is coverage in an integrated environment representative of a real small-to-midsize enterprise, not a set of isolated topic exercises.

Exam Domain	How This Project Covers It	NetAcad Reference
CCNA 2.1–2.2 — VLANs and interswitch connectivity	Six departmental VLANs, 802.1Q trunking, and inter-VLAN routing forming the segmented foundation	CCNA SRWE Modules 3–4
CCNA 2.3 — Spanning Tree and Layer 2 protections	PortFast, BPDU Guard, port security, and unused-port protection on the access layer	CCNA SRWE Modules 5, 11
CCNA 2.8–2.9 — Wireless client connectivity	Basic secure wireless to an access point, integrated with the wireless segment's DHCP scope and ACL policy	CCNA SRWE Modules 12–13
CCNA 4.3 — DHCP and DNS	Centralized DHCP addressing per segment and DNS internal name resolution, verified in the cross-service journey	CCNA SRWE Module 8; ENSA Module 5
CCNA 4.1 — Inside source NAT (NAT/PAT)	Outbound external connectivity via NAT and PAT, integrated with the segment ACL policy	CCNA ENSA Module 6
CCNA 4.5 — Syslog and NTP	Centralized Syslog collection with NTP synchronization, proven time-correct and shown to decorrelate under the induced NTP-failure test	CCNA ENSA Module 10
CCNA 5.5 — AAA concepts	RADIUS and AAA for centralized device-administration authentication, with SSH-enforced management and a documented local fallback	CCNA ENSA Module 3
CCNA 5.6 — Access control lists	Extended ACLs enforcing inter-segment access, verified in both directions and shown not to block RADIUS or DHCP	CCNA ENSA Module 4
CCNA 1.6 — IPv4 addressing and subnetting	The VLSM plan covering all segments, fixed in Week 1 as the shared reference	CCNA ITN Modules 11–12
CyberOps 1.7 — Defence-in-depth (secondary)	Segmentation, access control, authentication, hardening, and monitoring as layered controls that must interoperate, not stand alone	CyberOps Associate Module 4

Appendix B — Eight-Week Work Breakdown Structure

Each week represents 48 person-hours (4 students × 12 hours). Hours shown are the planned team total for that week. The Gate column defines what must be demonstrably working before the team proceeds; a missed gate is the project manager's trigger to re-plan rather than to carry the deficit forward.

Wk	Focus	Tasks	Hrs	Gate — must be true to proceed
1	Requirements, addressing, and the build standard	- Analyse the scope; produce a requirements list and a sponsor question log - Design the logical and physical topology and the complete VLSM addressing and segmentation plan - Write the build standard — naming, management access, hardening baseline, and per-service settings every device must conform to - Create the repository and shared reference documents - Present the addressing plan and build standard at the sponsor design review	48	SPONSOR DESIGN REVIEW. Addressing and segmentation plan and the build standard approved and agreed by all four members; repository created with all members committing.
2	Segmented topology build and capability spike	- CAPABILITY SPIKE: confirm RADIUS, wireless security, Syslog, and NTP behaviour in this Packet Tracer version - Build the topology in Packet Tracer to the build standard - Deploy VLANs, 802.1Q trunking, and inter-VLAN routing across all six segments - Verify intra-segment connectivity and inter-VLAN routing - Confirm every device conforms to the naming and baseline portions of the build standard	48	Base segmented topology functional; inter-VLAN routing confirmed working across all segments; capability spike complete.
3	Centralized services	- Deploy DHCP with a correct scope, gateway, and DNS option per segment - Deploy DNS for internal name resolution - Verify hosts in each segment receive the correct address and resolve names - Deploy NTP so that time is consistent across devices ahead of Syslog - Record every service against the build standard	48	DHCP and DNS functional across the appropriate segments; hosts receive correct per-segment addressing and resolve internal names.
4	Authentication and secure administration	- Deploy the RADIUS server and configure AAA for device administration - Verify a named administrative login authenticates via RADIUS - Enforce SSH as the management transport and disable unencrypted access	48	RADIUS authentication functioning for at least one administrative login; SSH replacing unencrypted management access on every device.

Wk	Focus	Tasks	Hrs	Gate — must be true to proceed
		- Document the local-authentication fallback and the intended RADIUS design in case of partial integration - Confirm management-access settings conform to the build standard		
5	Access control, external connectivity, and cross-training	- Deploy extended ACLs between segments per the segmentation plan, reviewed before application - Verify each access relationship in both the permitted and the denied direction - Deploy NAT and PAT for outbound external connectivity and verify it - Confirm the ACL layer does not block RADIUS, DHCP, or DNS - CROSS-TRAINING ROTATION: each member configures or verifies a component built by another member	48	ACLs enforcing intended relationships in both directions; outbound connectivity functional; each member has worked outside their own component.
6	Monitoring, wireless, and access-layer hardening	- Deploy Syslog collection and confirm entries arrive with NTP-correct timestamps - Deploy secure wireless to the access point and integrate it with the wireless segment's DHCP and ACL policy - Deploy port security, PortFast, BPDU Guard, and unused-port protection on access-layer switches - Verify a wireless client associates, authenticates, and is subject to its segment's policy - Confirm hardening settings conform to the build standard	48	Syslog receiving time-correct entries; secure wireless client connecting; access-layer hardening deployed to the build standard.
7	Integration, induced-failure testing, and non-author audit	- Run the cross-service end-to-end journey exercising wireless, RADIUS, DHCP, DNS, ACL, NAT, and Syslog together - Induce failures deliberately — break NTP; misorder an ACL to break RADIUS or DHCP — and demonstrate the integration failure and its repair - Document each integration conflict found and its resolution - NON-AUTHOR AUDIT: each member audits a component they did not build against the build standard, pass/fail per component - Resolve any nonconformance or integration defect surfaced by the audit	48	Cross-service journey passing end to end; induced-failure tests demonstrated and repaired; every component audited against the build standard by a non-author.
8	Delivery	Freeze the configuration; final regression run of the complete Appendix F test plan	48	FINAL DELIVERY. Full package complete and submitted; all four members rehearsed and able to explain the whole environment.

Wk	Focus	Tasks	Hrs	Gate — must be true to proceed
		- Export all device configurations and finalise both topology diagrams - Record the video demonstration including the cross-service journey and one induced-failure repair, all four members narrating - Build the slide deck and rehearse the presentation at least twice in full - Mock question-and-answer session with each member questioned on a component they did not build		

Appendix C — Team Roles and Hour Allocation

Four roles, 96 hours each. Roles describe accountability, not exclusive ownership: every student is expected to configure devices and to be able to explain any part of the build during the final presentation. The Project Manager role is a working role, not an administrative one — it carries a full technical workload in addition to coordination.

Role	Primary Accountability	Hrs
Project Manager and Network Architecture Lead	- Owns project coordination, the overall topology, the IPv4 addressing and subnet design, and the build standard - Ensures every component conforms to the approved architecture and the build standard - Chairs the weekly integration meeting, owns the Week 1 design review, and maintains the Appendix B gate checklist - Owns the sponsor relationship and final authority on the addressing plan and build standard sign-off	96
Switching and Routing Lead	- Owns VLANs, trunking, inter-VLAN routing, switching configuration, and Layer 2 protections - Owns access-layer configuration and the port-security, PortFast, BPDU Guard, and unused-port hardening - Accountable for the segmented foundation every other component builds on - Contributes the switching and hardening sections of the build standard	96
Network Services and Administration Lead	- Owns DHCP, DNS, RADIUS and AAA, SSH, Syslog, and NTP - Accountable for the RADIUS integration and its documented fallback, and for NTP being correct before Syslog depends on it - Owns the centralized-services sections of the build standard - Primary contributor to the cross-service journey design	96
Network Security, Integration, and Testing Lead	- Owns extended ACLs, NAT and PAT, wireless security, and infrastructure hardening review - Owns the cross-service test matrix, the induced-failure integration tests, and the integration-conflict log - Coordinates the Week 7 non-author audit and is custodian of the Appendix F evidence archive - Accountable for two-directional verification of every access relationship	96

Collaboration and version control

- All four members commit to the shared repository; because this is an integration project, a gap in one member's component breaks the whole, so a commit history showing work from only one or two members is a deliverable failure
- The addressing and segmentation plan and the build standard are the single sources of truth, fixed in Week 1; a setting on a device that departs from them is a defect whether or not it works
- Because Packet Tracer files cannot be merged, only one member edits the master .pkt at a time; members build in copies and hand changes to the Project Manager for integration
- Components are integrated continuously rather than at the end — a service is tested against the components it depends on as soon as both exist, not deferred to Week 7
- The Week 5 cross-training rotation and the Week 7 non-author audit both require every member to work outside their own component, which is what makes the presentation requirement that any member can explain the whole environment achievable

Appendix D — Topology Specification and Device Inventory

A representative enterprise topology sized to make each component meaningful: six departmental VLANs so segmentation and access control matter, a distribution or core layer providing inter-VLAN routing, an edge router for external connectivity and NAT, servers for DHCP/DNS and for RADIUS/Syslog/NTP, a wireless access point, and end devices across the segments sufficient to demonstrate correct and blocked connectivity. It is sized for this scope — not expanded with additional sites or redundancy — but it is large enough that the cross-service journey and the non-author audit are substantial rather than trivial. Device models are drawn from the standard Packet Tracer 8.2 palette; substitutions are documented.

Qty	Packet Tracer Device	Role in the Topology
2	ISR 4331 Router	Edge and core routing — one providing external connectivity, NAT/PAT, and the internet boundary; one supporting the routed core between segments
2	Catalyst 3560 Multilayer Switch	Distribution layer — inter-VLAN routing for all six segments and the enforcement point where inter-segment ACLs are applied
4	Catalyst 2960 Switch	Access-layer switches carrying the departmental VLANs, with port security, PortFast, BPDU Guard, and unused-port protection applied to the build standard
1	Server (DHCP/DNS)	Centralized addressing and internal name resolution across the appropriate segments
1	Server (RADIUS/AAA)	Centralized administrative authentication for device management — the component most likely to conflict with the ACL layer, and the subject of test T-11
1	Server (Syslog/NTP)	Centralized time-correct logging; the NTP source whose deliberate failure decorrelates the Syslog timeline in the induced-failure test
1	Wireless Access Point	Secure wireless connectivity for the Wireless segment, integrated with that segment's DHCP scope and ACL policy
1	Server (internal resource)	A protected Finance/Servers resource used as the permit/deny target in the two-directional segmentation tests
10	PC / wireless client	End devices across all six segments — at least one per segment plus a wireless client — providing the sources and destinations for functional, cross-service, and induced-failure testing

Design constraints the team must honour

- Every device must conform to the build standard; a working-but-nonconforming device is a defect, verified by the Week 7 non-author audit

- The addressing and segmentation plan is fixed in Week 1 and is the shared reference all members build against; changing it later requires re-baselining every dependent component
- Every access relationship must be proven in both directions — the permitted flow succeeds and the denied flow is blocked
- The ACL layer must be shown not to block RADIUS, DHCP, or DNS — the characteristic integration failure this project exists to prevent
- NTP must be correct before Syslog is relied upon; the induced NTP-failure test must demonstrate the timeline decorrelating and then recovering
- Wireless clients must be subject to the same DHCP and ACL policy as wired clients in their segment; a wireless client that bypasses segment policy is a defect
- Each component is capped at functional CCNA depth; depth belonging to a sibling scope (controller wireless, role-based AAA, SOC correlation, exhaustive ACL verification, multi-area OSPF) is out of scope and must not be pursued at the expense of integration
- SSH must replace unencrypted management access on every device; RADIUS must have a documented local fallback

Appendix E — Addressing and Segmentation Plan

The addressing plan below is the Week 1 reference every member builds against. Each department occupies its own VLAN and subnet so that segmentation and access control are meaningful, the server segment hosts the centralized services, and the management segment is separated from user traffic. The team finalises host-level detail within each allocation but the segment boundaries are fixed, because every downstream component — DHCP scopes, ACL rules, wireless policy — depends on them. The instructor has confirmed the team should use the flat /24 addressing as provided in this appendix rather than re-deriving it.

VLAN / Segment	VLAN ID	Subnet	Purpose and Policy
IT and Admin	10	10.20.10.0/24	IT and network administration; permitted management reach where the build standard specifies.
Finance	20	10.20.20.0/24	Protected financial resources; reachable only from permitted segments, denied from General Staff and Wireless.
General Staff	30	10.20.30.0/24	General user workstations; internet via NAT, denied to Finance and Management.
Servers	40	10.20.40.0/24	DHCP, DNS, RADIUS, Syslog, and NTP; reachable for the services each segment needs and no more.
Management	99	10.20.99.0/24	Device management interfaces; SSH-only, reachable from IT and Admin per the build standard.
Wireless	50	10.20.50.0/24	Wireless clients; DHCP-addressed and subject to the same ACL policy as wired users in an equivalent role.
Inter-VLAN / edge links	—	10.20.255.0/24	Point-to-point links and the NAT outside boundary to the simulated internet.
Simulated external	—	203.0.113.0/24	RFC 5737 documentation range; the outbound NAT/PAT destination that validates external connectivity.

Appendix F — Acceptance Criteria and Test Plan

This table is the definition of done. Each test must be executed and evidenced before the final presentation; the completed table, with evidence attached, is itself a graded deliverable. A test that cannot be made to pass must be documented as a known limitation with root-cause analysis — an honest, well-analysed failure scores higher than an undocumented one.

ID	Test Procedure	Expected Result	Evidence Required
T-01	A host in each VLAN attempts to reach both its intended and its unintended resources	Intended access succeeds and unintended access is blocked, consistent with the ACL plan, in both directions	Screenshot per segment showing the permitted success and the denied block, cross-referenced to the ACL plan
T-02	A host requests an address via DHCP in each segment	The host receives the correct address, gateway, and DNS server for its segment	Host addressing detail captured per segment
T-03	A host resolves an internal hostname via DNS	The hostname resolves to the correct address	Resolution capture from a client
T-04	An administrator authenticates to a network device for management access	RADIUS authenticates the login and the session is established over SSH; an unencrypted attempt is refused	Successful RADIUS-over-SSH login plus a refused unencrypted attempt
T-05	An internal host initiates outbound connectivity to an external address	NAT and PAT translate correctly and connectivity succeeds	Connectivity result plus the translation table entry
T-06	A wireless client connects to the access point and is subjected to its segment policy	The client associates and authenticates securely, receives DHCP, and is permitted or denied per its segment's ACL exactly as a wired client would be	Association and addressing capture plus a permit and a deny from the wireless client
T-07	A device on an access port exceeds the configured port-security address limit	The port takes the configured violation action	Port state and violation action captured
T-08	A device is connected to a port identified as unused	The port is administratively disabled and the device receives no connectivity	Port state and a failed connectivity attempt
T-09	A monitored device generates log activity	The entries appear on the Syslog server with correct, NTP-synchronized timestamps	Syslog capture showing entries with consistent timestamps across devices
T-10	CROSS-SERVICE JOURNEY: a wireless client associates, authenticates via RADIUS-backed policy, receives DHCP, resolves DNS, reaches a permitted resource, is denied a restricted one, reaches the internet via NAT, and appears in Syslog	Every stage succeeds or is denied exactly as designed, in a single unbroken journey	A staged capture of each step of the journey, demonstrating the components work together
T-11	INDUCED FAILURE: introduce an ACL that blocks RADIUS (or DHCP) traffic, observe the break, then correct the ordering	The integration failure manifests (authentication or addressing fails), is correctly diagnosed, and is repaired	Before, broken, and after captures with the integration-conflict log entry explaining the cause
T-12	INDUCED FAILURE: stop or misconfigure NTP and generate log activity across devices	Syslog timestamps decorrelate across devices; after NTP is restored, timestamps realign	Syslog captures before, during, and after the NTP failure showing the timeline decorrelate and recover
T-13	Verify DHCP, DNS, RADIUS, and NAT continue to function with the full ACL policy applied	No centralized service is broken by the access-control layer	Functional confirmation of each service with all ACLs in place
T-14	NON-AUTHOR AUDIT: a member audits a component they did not build against the build standard	The component conforms to the standard, or nonconformance is identified	Signed audit checklist per component with any

ID	Test Procedure	Expected Result	Evidence Required
		and corrected; the audit produces a pass/fail per component	nonconformance and its correction
T-15	Run 'show ip ospf/route' and 'show ip access-lists' across the environment to confirm build-standard consistency	Configuration is consistent with the build standard across every device of a kind; ACL hit counters reflect the test traffic	CLI captures compared against the build standard, plus hit-counter evidence

Appendix G — Grading Rubric

Weights total 100%. The rubric is published with the scope so that teams can self-assess before submission.

Component	Wt	What full marks looks like
Segmented and routed foundation	15%	All six VLANs present and correctly interconnected; trunking correct; inter-VLAN routing working; the .pkt opens without error with every device named per the build standard.
Centralized services (DHCP/DNS/NAT/Syslog/NTP)	15%	Correct per-segment DHCP, working DNS, functioning NAT/PAT, and centralized Syslog with NTP-correct timestamps — each functional and each conforming to the build standard.
Authentication and secure administration	15%	RADIUS authenticating administrative login; SSH enforced and unencrypted access refused; the local fallback documented; management access conforming to the build standard.
Access control and segmentation enforcement	15%	Extended ACLs enforcing the intended relationships, verified in both directions with hit-counter evidence, and shown not to block RADIUS, DHCP, or DNS.
Wireless and access-layer hardening	10%	Secure wireless to the access point integrated with segment DHCP and ACL policy; port security, PortFast, BPDU Guard, and unused-port protection deployed to the build standard.
Integration, interoperation, and consistency	25%	The cross-service journey passes end to end; both induced-failure tests are demonstrated and repaired with a documented conflict log; every component passes the non-author audit against the build standard; the environment works as one coherent, consistent whole. This is the heaviest component because integration is the project.
Final presentation and video	5%	All four members present and able to answer for any component including ones they did not build; the live demonstration works; the video shows the cross-service journey and an induced-failure repair.

Appendix H — Stretch Goals

The core scope in the table above is sized to occupy a competent four-student team for the full 384 hours. Stretch goals are strictly optional and are the mechanism by which a strong team distinguishes itself. No stretch goal may be attempted until every acceptance-criteria test in Appendix F passes — an incomplete core with an impressive stretch goal scores below a complete core alone.

- Extend the ACL design to a more granular, least-privilege ruleset per segment pair, and prove the tighter policy with the same two-directional verification
- Add a guest wireless network fully isolated from all internal segments, and prove the isolation by attempting internal access from a guest client
- Automate a build-standard conformance check — a short script or checklist that reads exported configurations and flags any device that departs from the standard — and run it as the non-author audit
- Add a second small site connected over a simulated link and demonstrate that the centralized services (DHCP relay, DNS, RADIUS, Syslog) extend to it correctly, analysing what changes as the enterprise grows
- Document a short, evidenced plan for how this design would scale with redundancy in a future phase — redundant links, a first-hop redundancy protocol, and a second services server — without redesign

- Build an integration troubleshooting field guide mapping each characteristic cross-service failure — wireless client with no address, RADIUS blocked by ACL, Syslog timestamps wrong — to its diagnostic sequence and root cause

Appendix I — Final Presentation and Video Demonstration

Final presentation (Week 8)

A 10-minute presentation to the sponsor and faculty — 6 minutes on the technical build and demonstration, and 4 minutes on the MDDDE and other required IT491 processes — per the instructor's confirmed timing for this presentation. All four students present. The panel will direct technical questions to individual students at random, so every member must be able to explain any part of the build — a team that can only answer through one spokesperson will be marked down.

- Technical build walkthrough and live demonstration against the acceptance criteria, including at least one deliberate failure/recovery scenario (6 min)
- The MDDDE and other required IT491 processes (4 min)

Video demonstration

A recorded walkthrough of 8–12 minutes, submitted with the final package. The video must show live Packet Tracer interaction with visible CLI output — narrated slides alone are not acceptable. Every team member must narrate a section of the video. The instructor has confirmed this video is expected as part of the final deliverable.

- A walkthrough of the six-segment enterprise, identifying each VLAN, the centralized services, and the enforcement points
- The cross-service end-to-end journey as the hero demonstration: one wireless client associating, authenticating, addressing, resolving, being permitted and denied, reaching the internet, and appearing in Syslog
- An induced-failure repair — an ACL misordered to block RADIUS, diagnosed and fixed — shown live
- The NTP-failure test showing Syslog timestamps decorrelate and then realign after NTP is restored
- A non-author audit confirming a component conforms to the build standard

Final submission package

- The Packet Tracer .pkt file, opening without errors in version 8.2 or later
- Exported running configurations for every configured device, as plain text
- Network topology diagram, logical and physical views
- Completed Appendix F test plan with evidence attached
- Project report incorporating the design rationale, addressing plan, and known limitations
- Presentation slide deck
- Video demonstration file or link
- Version control repository link with a commit history showing contribution from all four students